Design Thinking Report for Crazy 1-0s Game

Team #25, The Crazy Four

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April 6, 2026

Contents

1 Introduction **3**

2 Design Challenge **3**

3 Target Users **4**
3.1 Primary Users 4
3.2 Secondary Users and Stakeholders 4
3.3 Shared User Needs 4

4 Interview Plan **4**
4.1 Purpose of the Interviews 4
4.2 Interview Objectives 5
4.3 Sample Interview Questions 5
4.4 Rationale 5

5 Empathy Findings **6**
5.1 What Users Say 6
5.2 What Users Think 6
5.3 What Users Do 6
5.4 What Users Feel 6
5.5 Summary of Empathy Findings 6

6 Key Insights and Problem Definition **7**
6.1 Insight 1: Users Resist Overload More Than Learning Itself 7
6.2 Insight 2: Familiar Gameplay Lowers the Entry Barrier 7
6.3 Insight 3: Feedback Functions as a Learning Mechanism 7
6.4 Refined Problem Definition 7

7 Prototype Development **7**
7.1 Prototype 1: Preliminary Sketches and Rule Exploration 8
7.2 Prototype 2: Core Gameplay Wireframe 10
7.3 Prototype 3: Refined Interface Variation 11

8 User Feedback and Comparison **12**
8.1 Comparison Across the Three Prototypes 12
8.2 Feedback from Prof. Rapoport 13
8.3 Feedback from the Instructor 13

9 Design Iteration **14**

10 Impact on the Broader Project **15**

11 Reflection **15**

1 Introduction

This document presents the Design Thinking process used to support the development of the *Crazy 1-0s / Crazy Tens* project. The project aims to create a digital card game that introduces users to dozenal concepts through interactive gameplay. Since the project combines educational objectives with entertainment-oriented interaction, the design problem is not purely technical. In addition to implementing a working game system, the project must ensure that users can understand the rules, remain engaged, and perceive the learning elements as helpful rather than burdensome.

A Design Thinking approach was appropriate because the success of the project depends heavily on the user experience. Even if the system is functionally correct, it may still fail if first-time users find the game confusing, intimidating, or excessively instructional. Accordingly, the purpose of this report is to document how user-centered investigation informed the design of the system before major design choices were finalized.

This report focuses on the stages of the Design Thinking process that are most relevant to the current phase of the project: understanding the problem context, identifying target users, planning and interpreting interviews, organizing empathy-based findings, defining the design problem, comparing early prototype directions, analyzing user feedback, and identifying the implications of these findings for the broader project.

2 Design Challenge

The central design challenge of this project is to introduce dozenal concepts in a form that is understandable, engaging, and accessible to first-time users. Most potential users are highly familiar with decimal representation and may have little to no prior exposure to base-12 notation. As a result, the project must do more than merely present alternative numeral representations. It must actively reduce the cognitive and emotional barriers associated with unfamiliar mathematical content.

This challenge is complicated by the dual nature of the project. On one hand, the system is intended to serve an educational purpose by helping users become more comfortable with dozenal ideas, numerical representation, and related mechanics. On the other hand, it must still function as a game. If the system places too much emphasis on instruction, it risks becoming tedious or intimidating. If it places too little emphasis on learning support, it may fail to achieve its educational objective.

The design challenge can therefore be stated as follows:

> **How might we design a digital card game that introduces dozenal concepts in a way that feels approachable, engaging, and not overwhelming to first-time users?**

This question guided the remainder of the Design Thinking process. It helped shift the focus from simply adding features to determining which design choices would most effectively support user understanding and sustained engagement.

3 Target Users

3.1 Primary Users

The primary users of the proposed system are students and other first-time players who are familiar with simple card-game mechanics but are unfamiliar with dozenal notation and related mathematical ideas. These users are especially important because they represent the intended learning audience of the system. Their responses are likely to reflect whether the game is successful as an introductory educational experience.

A second segment of the primary user group consists of casual players who enjoy rule-based games and are open to experimental or educational mechanics, provided that the experience remains intuitive and enjoyable. This group is useful because it helps evaluate whether the project can function as a game in its own right, rather than only as a teaching aid.

3.2 Secondary Users and Stakeholders

Secondary users and stakeholders include instructors, supervisors, teaching assistants, and evaluators who may view the system as an educational tool or as an example of user-centered design in software engineering. Although these users are not necessarily the main gameplay audience, their perspective is important when evaluating whether the learning objectives are clearly represented and whether the design decisions are justified.

3.3 Shared User Needs

Despite differences among these groups, several user needs were identified as common and significant:

- a clear and approachable introduction to the rules,
- gradual exposure to unfamiliar dozenal concepts,
- meaningful feedback when mistakes occur,
- a game structure that feels coherent and familiar,
- and a balance between educational value and enjoyment.

Recognizing these shared needs helped frame the project in terms of user experience rather than implementation convenience.

4 Interview Plan

4.1 Purpose of the Interviews

The interview stage was intended to help the team understand how potential users approach unfamiliar games, how they respond to educational mechanics, and what kinds of guidance they find useful when learning new rules. The interviews were not designed to confirm that

the existing concept was already correct. Rather, their purpose was to gather evidence about user expectations, frustrations, and preferences so that the design could be refined accordingly.

4.2 Interview Objectives

The interviews were structured around the following objectives:

1. To understand users' prior familiarity with card games and rule-learning processes.
2. To investigate how users react to mathematical or educational content within a game.
3. To identify what makes a game feel approachable or intimidating.
4. To determine what form of tutorial, hint, or feedback users find most useful.
5. To reveal design risks that might not be obvious from the development team's perspective.

4.3 Sample Interview Questions

To support these objectives, the interview questions were designed to be open-ended and exploratory. Representative questions included the following:

1. What kinds of card games or turn-based games are you already familiar with?
2. When learning a new game, what helps you understand the rules most effectively?
3. How would you respond if a game introduced a mathematical idea that you had not previously encountered?
4. What would make an educational game feel enjoyable rather than forced?
5. What types of hints or tutorials do you find helpful, and what types do you tend to ignore?
6. If you made a mistake because of an unfamiliar rule, what sort of feedback would you want from the system?
7. Would you prefer to learn most rules before starting, or gradually during gameplay?

4.4 Rationale

These questions were selected because they allow users to describe their own experiences, habits, and frustrations rather than merely reacting to a yes-or-no proposal. This approach makes it easier to identify patterns in user behaviour and expectation, which is the primary purpose of the early Design Thinking stages.

5 Empathy Findings

Following the interviews, the observed responses were interpreted through an empathy-based framework. This helped organize the findings into patterns related to what users say, think, do, and feel when interacting with a new game concept.

5.1 What Users Say

Users generally reported that they were comfortable with familiar card-game structures, particularly when the core interaction involved recognizable actions such as playing, drawing, and matching cards. However, they also indicated that a system may become discouraging if too many new rules are introduced at once. Several users expressed a preference for concise and contextual explanations rather than long introductory tutorials.

5.2 What Users Think

The interviews suggested that many users implicitly view dozenal content as unfamiliar and potentially difficult, even when they are curious about it. There was also evidence that some users associate educational games with lower entertainment value. This indicates that the system must establish trust quickly by demonstrating that it is still enjoyable as a game, even while it introduces learning-oriented mechanics.

5.3 What Users Do

Users tend to rely on familiar patterns when approaching a new system. In practice, this means that they often search for visual cues, compare new rules to rules they already know, and pause when they are uncertain whether an action is valid. If explanations are too dense or separated from the moment in which confusion occurs, users may skip them and instead attempt to proceed by trial and error.

5.4 What Users Feel

The emotional responses inferred from the interviews included a mixture of curiosity, hesitation, and concern about making mistakes. Users may be interested in the novelty of a different numeral system, but they are less likely to persist if the game makes them feel incompetent early on. By contrast, a design that frames mistakes as informative and recoverable may help transform uncertainty into engagement.

5.5 Summary of Empathy Findings

Overall, the empathy findings indicate that the project must balance three requirements carefully:

- familiarity, so that users are willing to begin;
- support, so that users can understand what is happening;

- and novelty, so that the educational purpose of the project remains meaningful.

6 Key Insights and Problem Definition

Several important insights emerged from the interview and empathy stages.

6.1 Insight 1: Users Resist Overload More Than Learning Itself

The evidence suggests that users are not necessarily opposed to learning dozenal concepts. Rather, they are more likely to resist an experience that appears dense, complicated, or inaccessible at the beginning. This means that the primary obstacle is not the content itself, but the manner in which it is introduced.

6.2 Insight 2: Familiar Gameplay Lowers the Entry Barrier

A recognizable card-game structure provides an important scaffold for the learning experience. By embedding new ideas within a familiar framework, the system can reduce uncertainty and encourage users to continue interacting with the game even when unfamiliar elements appear.

6.3 Insight 3: Feedback Functions as a Learning Mechanism

In this project, system feedback is not merely a usability feature. Since the game introduces unfamiliar representations and rules, feedback also serves as an instructional tool. Well-timed explanations can transform errors into learning opportunities, while unclear rejection messages may increase frustration and confusion.

6.4 Refined Problem Definition

Based on these insights, the design problem was refined into the following statement:

> **First-time users require a game experience that introduces dozenal concepts gradually through familiar gameplay, contextual guidance, and low-pressure feedback, because abrupt complexity and poorly timed explanations can make the system feel confusing rather than educational.**

This refined definition provided a more concrete basis for comparing prototype directions and interpreting later feedback.

7 Prototype Development

This section summarizes the prototyping stage of the Design Thinking process. Based on the materials developed by the team, three prototypes are identified in the progression of the project. The first prototype consisted of early paper sketches and handwritten rule drafts.

The second prototype presented a clearer wireframe of the main gameplay screen. The third prototype further refined the same gameplay layout by adding more interface completeness and peripheral controls. Together, these three prototypes illustrate how the design evolved from broad concept exploration to a more coherent and defensible interface direction.

7.1 Prototype 1: Preliminary Sketches and Rule Exploration

The first prototype was a low-fidelity paper prototype used to externalize the earliest design ideas for the game. At this stage, the team was still determining how the educational and gameplay components should coexist within one system. As a result, the prototype included multiple exploratory sketches rather than one finalized layout.

Prototype 1 included rough drawings of the opening interaction, number-base selection, game mode selection, and the general game board layout. One sketch explored the idea that the player should explicitly choose between decimal and dozenal before starting the game. Another sketch proposed an early menu for selecting between two-player and multiplayer modes. Additional sketches showed the rough placement of the draw pile, discard pile, player hand, and rule display area. The prototype also included handwritten notes describing two-player rules, wild card behaviour, arithmetic operation mechanics, options, timing concepts, and point calculation ideas.

This prototype was intentionally incomplete. The interface was monochrome and unpolished, the rule set was still changing, and many features were presented only as rough notes. Some parts of the design were cluttered, and the connection between educational mechanics and core gameplay was not yet fully resolved. However, this was acceptable because the purpose of Prototype 1 was not to test visual polish. Its role was to make the early concept concrete enough that the team could identify missing pieces, inconsistencies, and overly ambitious ideas.

Prototype 1 was especially useful for surfacing the following design questions:

- how number-base selection should be presented to the player;
- whether multiple game modes should be supported;
- where rules and options should appear so that they remain accessible without dominating the screen;
- how arithmetic interactions should be triggered by cards;
- and how much information should be shown at one time without overwhelming the user.

For these reasons, Prototype 1 served primarily as a concept exploration tool. It allowed the team to externalize assumptions and begin narrowing the design space.

Screen
↳ selected Dozenal

Crazy 10s! (Dozenal)

→ 2-player Multiplayer (3-5)

Crazy 10s!

Dozenal

Game Rules

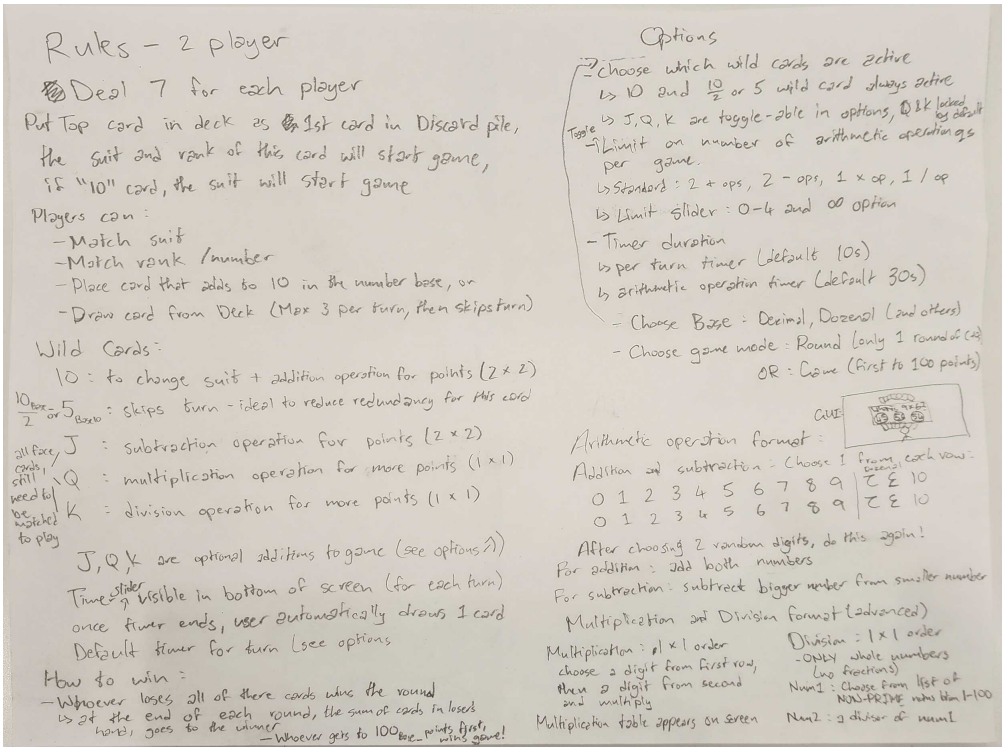

Figure 2: Prototype 1 - Preliminary Sketches (Part 2)

7.2 Prototype 2: Core Gameplay Wireframe

The second prototype marked a shift away from broad concept exploration and toward a more focused representation of the in-game user experience. In this version, the design concentrated on the main gameplay screen and the information that would need to remain visible during play.

Prototype 2 presented a simplified wireframe of the central game board. It displayed the player's score, the opponent's score, the target score, and the selected base at the top of the screen. Below this, the layout clearly separated the opponent's hand, the draw pile, the discard pile, and the player's hand. At the bottom, the essential interaction buttons were reduced to direct and recognizable actions such as drawing a card or playing a card.

Compared with Prototype 1, this prototype was far more coherent as an interface. The major gameplay regions were easier to identify, the hierarchy of information was clearer, and the basic flow of a turn became more understandable. Instead of trying to represent every possible feature or rule at once, Prototype 2 focused on the recurring interaction pattern of the game and the minimum information needed to support it.

Prototype 2 was useful for evaluating whether a first-time user could quickly understand the layout of play. In particular, it helped the team reason about whether the score information was visible enough, whether the piles and hands were visually distinct, and whether the interface emphasized the player's most important actions in a sufficiently direct way.

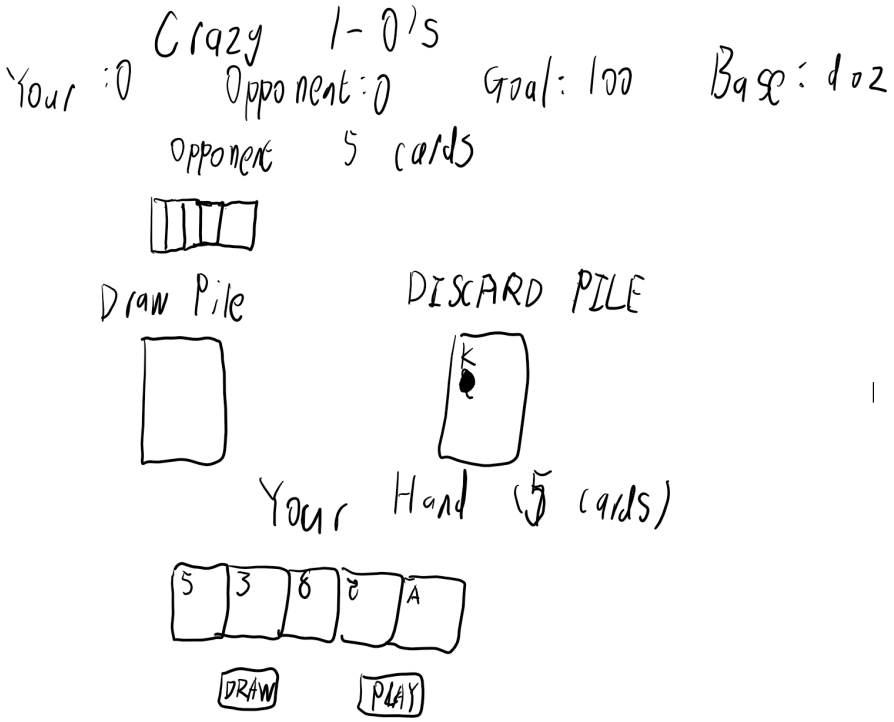

Figure 3: Prototype 2 - Core Gameplay Wireframe

7.3 Prototype 3: Refined Interface Variation

The third prototype further refined the direction established by Prototype 2. It preserved the core gameplay layout but added greater interface completeness by introducing peripheral controls and suggesting how auxiliary functions could be accessed in a more finished product.

In this version, the main board still centered on the same elements as in Prototype 2: the opponent area, the draw pile, the discard pile, the player's hand, and the main control buttons. However, extra interface elements were added around the sides of the screen, indicating how secondary actions or supporting features could be integrated without disrupting the core gameplay arrangement.

Prototype 3 is important because it shows that the design had progressed beyond a simple wireframe into a more product-oriented interface concept. It demonstrated that the layout developed in Prototype 2 was stable enough to support additional functionality. Rather than replacing the earlier design, Prototype 3 extended it, suggesting a direction for how the game could feel more complete while preserving clarity.

This prototype therefore represented a refinement stage rather than a conceptual restart. It helped the team consider not just the play area itself, but also the broader question of how the interface could remain expandable and usable as more features were incorporated.

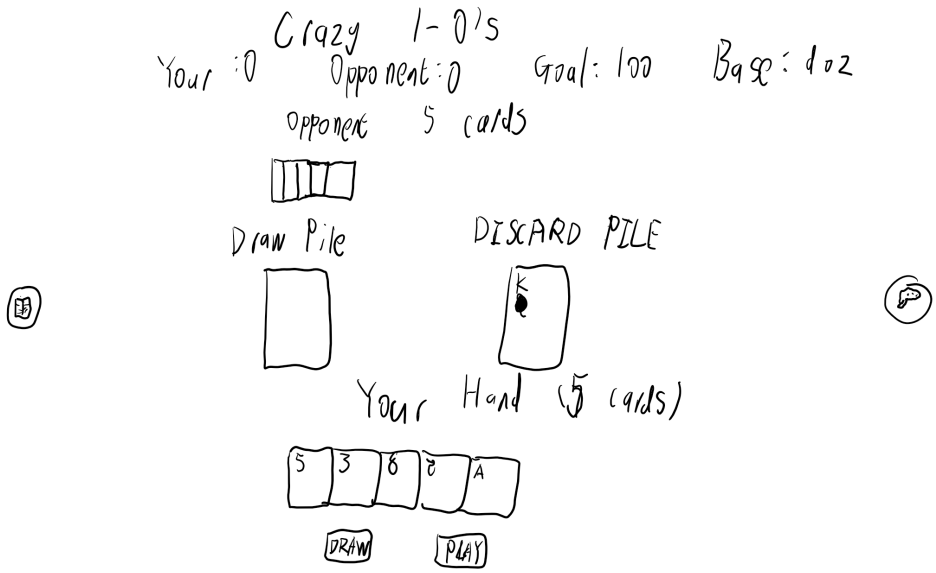

Figure 4: Prototype 3 - Refined Interface Variation

8 User Feedback and Comparison

8.1 Comparison Across the Three Prototypes

The three prototypes demonstrate a gradual narrowing and strengthening of the design direction.

Prototype 1 emphasized exploration. It was broad, speculative, and useful for identifying the range of possible mechanics, modes, and educational interactions. However, it also revealed the danger of trying to communicate too many ideas simultaneously.

Prototype 2 emphasized focus. It took the most important recurring gameplay elements and organized them into a clearer wireframe that was easier to understand and more suitable for user-centered evaluation.

Prototype 3 emphasized refinement. It retained the successful structure of Prototype 2 while indicating how additional controls and supporting features could be incorporated into a more complete interface.

Taken together, the three prototypes show a natural progression from concept exploration, to gameplay wireframing, to interface refinement. This progression reflects the convergence phase of Design Thinking, in which early ideas are gradually reduced and improved until a more defensible direction emerges.

8.2 Feedback from Prof. Rapoport

Feedback from Prof. Rapoport contributed primarily to the refinement of game rules, educational mechanics, and the relationship between numeral bases and gameplay. His comments helped identify areas where the rule system needed stronger consistency and where some mechanics could be simplified.

One major area of feedback concerned the relationship between decimal and dozenal play. The game was expected to support both bases, and several related rule details needed to be made explicit, such as the number of cards dealt, the number of wild cards, and the point target associated with each base. This reinforced the need for base selection to be treated as a meaningful game-setting decision rather than as a cosmetic label.

Another important part of the feedback addressed wild cards and arithmetic triggers. The comments suggested that the arithmetic system should be simplified and made more systematic. Instead of relying on random operation choice or unnecessarily complicated trigger logic, the arithmetic effects should be more directly linked to recognizable card categories or suits. This advice reduced ambiguity and helped move the design toward a structure that would be easier for players to learn and for the team to implement consistently.

The feedback also emphasized that typed answers would be more appropriate for arithmetic questions than more complicated multiple-choice generation. In addition, optional educational support resources, such as static addition or multiplication tables, were suggested as a better form of support than external media such as videos.

Overall, the feedback from Prof. Rapoport was valuable because it pushed the project toward stronger rule consistency, simpler educational mechanics, and a more coherent relation between gameplay and learning.

8.3 Feedback from the Instructor

The instructor feedback focused more heavily on implementation completeness, visual clarity, and practical usability. While the earlier prototypes were valuable for concept development, the feedback on the Rev0 direction emphasized that the system should move closer to a complete and credible implementation.

One major point was that the interface and implementation needed to progress beyond placeholder-level representation. The feedback specifically noted that actual card designs should be used rather than overly abstract substitutes, and that certain visual or unicode issues required correction. This reinforced the importance of making the game appear understandable and polished enough for users to interpret game state correctly.

Additional feedback highlighted missing or incomplete system features, such as authentication, timing options, support for multiple simultaneous games, and leaderboard or database integration. Although not all of these items were necessarily the highest priority, their identification showed that the project would ultimately be judged not only as a concept, but also as a functioning software system.

The instructor also stressed the importance of testing and usability evaluation. Since the

project involves a rule-based game with educational mechanics, the team was encouraged to think carefully about edge cases and to ensure that game rules would always be enforced correctly. This further confirmed that usability and correctness are closely linked in the context of this project.

Taken together, the instructor feedback pushed the design from conceptual refinement toward implementation realism. It encouraged the team to think not only about what the interface should contain, but also about whether the overall system would be complete, understandable, and robust enough to support real use.

9 Design Iteration

The progression from Prototype 1 to Prototype 2 and then to Prototype 3 illustrates the iterative nature of the Design Thinking process in this project.

The first prototype made it possible to externalize broad ideas about gameplay, arithmetic learning, base selection, and rule variation. However, it also showed that the system risked becoming too complicated if too many ideas were presented at once.

The second prototype addressed this by reducing the interface to a clearer gameplay wireframe centered on the main play experience. This made the recurring structure of the game easier to understand and provided a stronger basis for discussing usability.

The third prototype then built on this improvement by extending the same basic layout into a more complete interface concept. Instead of redesigning the system again, it refined the established structure and suggested how more functionality could be added without losing clarity.

The two groups of external feedback further strengthened this process. Prof. Rapoport's comments improved rule consistency and educational mechanics, while the instructor feedback emphasized implementation completeness, testing, and usability. Together, these responses helped the team move from early conceptual enthusiasm toward a design that was clearer, more practical, and more defensible.

The final design direction that emerged from this process can be summarized as follows:

- maintain a familiar and readable card-game layout;
- support both decimal and dozenal in a more systematic way;
- simplify arithmetic interactions so that they remain educational without becoming confusing;
- present core game information clearly during play;
- and refine the interface incrementally rather than introducing unnecessary complexity too early.

This iterative development path demonstrates that prototyping was not merely a visual exercise. It was a way of identifying uncertainty, collecting feedback, and progressively

improving both the gameplay concept and the interface structure of the project.

10 Impact on the Broader Project

The Design Thinking process influenced the broader project in several important ways.

First, it shifted the design perspective from feature-centered reasoning to user-centered reasoning. Instead of focusing only on what mechanics could be implemented, the team was prompted to consider what users would understand, what they would hesitate over, and what would motivate continued engagement.

Second, it strengthened the rationale for several project priorities. The need for onboarding support, progressive learning assistance, and clear explanatory feedback became more significant after the user-centered stages of the process. These issues are not merely implementation details; they are central to whether the project can achieve its intended purpose.

Third, the process improved traceability. Design choices could now be linked back to observed or inferred user needs rather than justified solely by team preference. This strengthens the overall coherence of the project by connecting design decisions to concrete evidence gathered during early investigation.

For these reasons, the Design Thinking process served not as a separate exercise, but as a foundation that informed later requirements, interface decisions, and implementation priorities.

11 Reflection

The Design Thinking process provided several important lessons.

First, it demonstrated that educational software cannot be designed effectively from the development team's perspective alone. Even when the intended learning objective appears clear internally, users may interpret the same design very differently depending on their background, confidence, and expectations.

Second, it showed that uncertainty in the early stages of design is not necessarily a weakness. Questions regarding how visible the educational elements should be, how strong the feedback should be, and how much novelty users can tolerate proved valuable because they led to more focused inquiry and more meaningful prototype comparison.

Third, the process reinforced the importance of iteration. Initial design directions were useful starting points, but they became more defensible only after being examined through interview-based evidence, empathy analysis, and prototype feedback. This suggests that effective software design is not solely a matter of building a functional system, but of progressively refining a system so that it aligns with the needs and expectations of its intended users.

In future software engineering work, this experience will remain relevant wherever user understanding, confidence, and engagement are critical to project success. It illustrates that

human-centered design is not an optional addition to development, but an essential part of creating software that is both usable and meaningful.